

LAB-1

1. Write a Python program that accepts the users first and last name and prints them in reverse order with a space between them.
- 2 .Write a Python program to sum the first n positive integers.
3. Write a program which will find all such numbers which are divisible by 7 but are not a multiple of 5, between 2000 and 3200 (both included).

4. Write a program that calculates and prints the value according to the given formula:
 $Q = \text{Square root of } [(2 * C * D)/H]$

Following are the fixed values of C and H:

C is 50. H is 30. D is the variable whose values should be input to your program in a comma-separated sequence.

Example

Let us assume the following comma separated input sequence is given to the program:

100,150,180

The output of the program should be: 18,22,24

Hints:

If the output received is in decimal form, it should be rounded off to its nearest value (for example, if the output received is 26.0, it should be printed as 26)

In case of input data being supplied to the question, it should be assumed to be a console input.

5. Write a program to print a right angle triangle with alphabets.(instead of '*', Alphabet).

Hint: How to do it for

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LAB-2

1. Write a program to create a class for student data (student ID, name, marks, and percentage). Implement a method to calculate the percentage and grade (A, B, C, D) based on the percentage. Take the values as input at runtime.

2. Write a program to demonstrate multiple inheritance where one class represents a person and another represents a teacher. The derived class should inherit both functionalities and display teacher and person details.

3. Write a program to validate a phone number using regular expressions. Ensure the number follows the format: XXX-XXX-XXXX.

4. Write a program that accepts a list of student scores and performs the following tasks:

- a. Sort the list in ascending order.
- b. Identify and remove duplicate scores.
- c. Calculate the average score.

5. Write a program to simulate a simple library system. Implement classes for Book and Member. Allow members to borrow and return books, and track which books are currently borrowed.

LAB-3

1. In the regex created from `r(\d\d\d)-(\d\d\d-\d\d\d\d)`, what does group 0 cover?

Group 1? Group 2?

2. Clean up dates in different date formats (such as 3/14/2015, 03-14-2015, and 2015/3/14) by

replacing them with dates in a single, standard format.

3. How would you write a regex that matches a number with commas for every three digits? It

must match the following:

- '42'
- '1,234'
- '6,368,745'

but not the following:

- '12,34,567' (which has only two digits between the commas)
- '1234' (which lacks commas)

4. How would you write a regex that matches the full name of someone whose last name is

Nakamoto? You can assume that the first name that comes before it will always be one word

that begins with a capital letter.

The regex must match the following:

- Satoshi Nakamoto
- Alice Nakamoto
- RoboCop Nakamoto

but not the following:

- santoshi Nakamoto (where the first name is not capitalized)
- Mr. Nakamoto (where the preceding word has a nonletter character)
- Nakamoto (which has no first name)
- Satoshi nakamoto (where Nakamoto is not capitalized)

5.Strong Password Detection Write a function that uses regular expressions to make sure the password string it is passed is strong. A strong password is defined as one that is at least eight characters long, contains both uppercase and lowercase characters, and has at least one digit.

You may need to test the string against multiple regex patterns to validate its strength.

LAB-4

- 1.Import the required Python libraries (pandas, numpy, matplotlib, seaborn).
- 2.Load the dataset into a pandas DataFrame.
- 3.Display the number of rows and columns in the dataset.
- 4.Print the column names of the dataset.
- 5.Display the first 5 rows of the dataset.
- 6.Check the data type of each column.
- 7.Identify the number of missing values in each column.
- 8.Fill missing values in numerical columns using the mean.
- 9.Fill missing values in categorical columns using the mode.
- 10.Calculate the mean for all numerical columns.
- 11.Calculate the median for all numerical columns.
- 12.Calculate the standard deviation for all numerical columns.
- 13.Find the minimum and maximum values for numerical columns.

LAB-5

- 1.Write a NumPy program to create a 3x3 matrix with values ranging from 2 to 10. Expected Output: `[[ 2 3 4] [ 5 6 7] [ 8 9 10]]`
2. Write a NumPy program to find common values between two arrays. Expected Output: `Array1: [ 0 10 20 40 60] Array2: [10, 30, 40] Common values between two arrays: [10 40]`

3. Write a NumPy program to find the set difference between two arrays. The set difference will return sorted, distinct values in array1 that are not in array2. Expected Output: Array1: [0 10 20 40 60 80] Array2: [10, 30, 40, 50, 70, 90] Set difference between two arrays: [0 20 60 80]

4. Write a NumPy program to compare two arrays using NumPy.
Array a: [1 2]

Array b: [4 5]
a > b [False False]
a >= b [False False]
a < b [True True]
a <= b [True True]

5. Multiply Array (3,4) Elements by 3

Write a NumPy program to create an array of (3, 4) shapes, multiply every element value by 3 and display the result array.

Expected Output:

Original array elements:

```
[[ 0 1 2 3]
 [ 4 5 6 7]
 [ 8 9 10 11]]
```

New array elements:

```
[[ 0 3 6 9]
 [12 15 18 21]
 [24 27 30 33]]
```

Lab-6: one-way Anova Question

1. A teacher wants to check whether three different teaching methods produce different studentscores.

Scores are:

Method A	Method B	Method C
78	85	92
82	88	95
85	90	89
80	87	94

Tasks:

1. State null and alternative hypotheses.
2. Calculate group means.
3. Calculate Total Sum of Squares (SST).
4. Calculate Sum of Squares Between (SSB).
5. Calculate Sum of Squares Within (SSW).
6. Construct ANOVA table.
7. Test at $\alpha = 0.05$.

8. Write conclusion.

2. Two-Way ANOVA Question

A researcher studies effect of:

Fertilizer Type (A, B)

Water Level (Low, High)

Crop yield data:

Low Water High Water

Fertilizer A 30, 32, 31 40, 42, 41

Fertilizer B 28, 29, 27 38, 37, 39

Tasks:

1. Identify factors.
2. Test main effects.
3. Test interaction effect.
4. Write interpretation.